

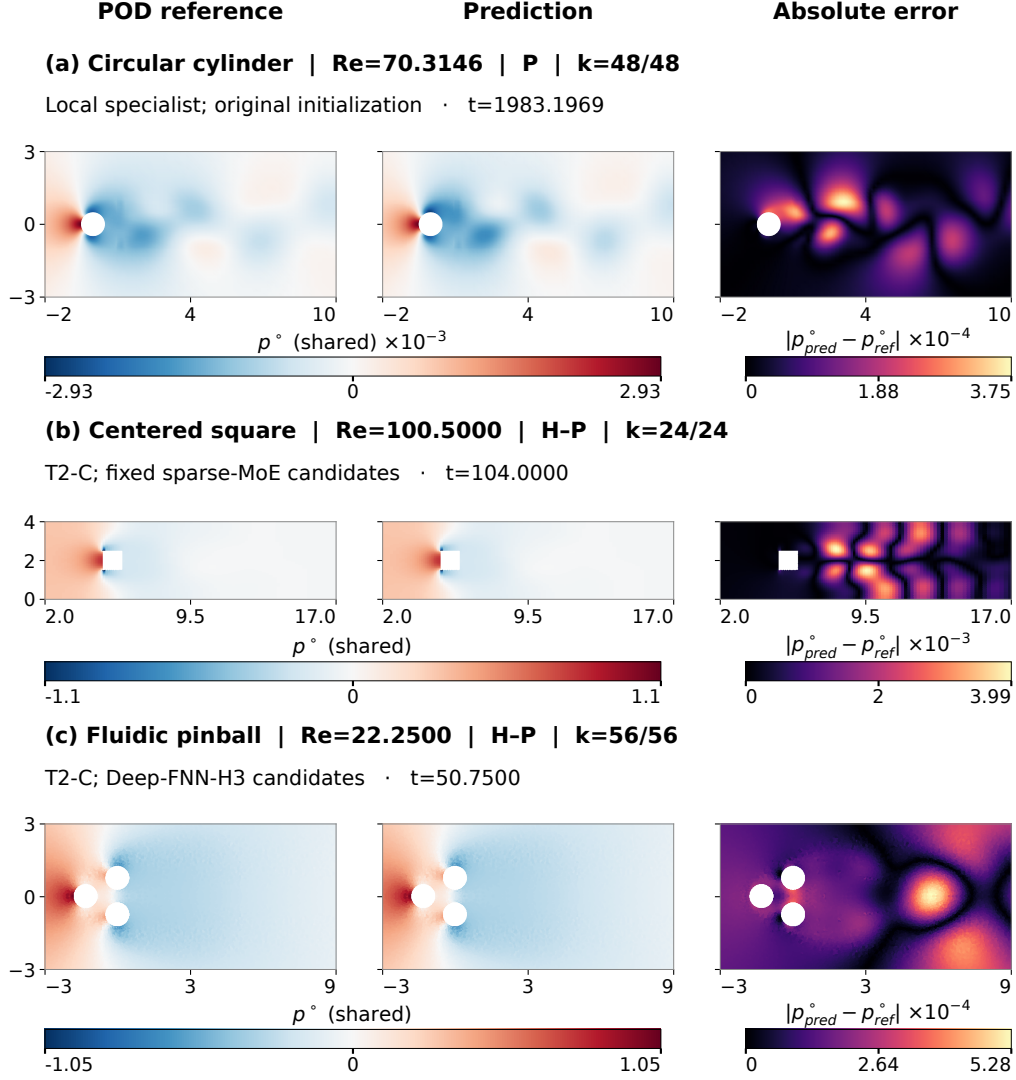


Figure 1: Representative pressure fields across three flow benchmarks. Columns show the POD-reconstructed reference, the case-specific prediction, and pointwise absolute error. Circular uses the original-initialization Periodic sparse-MoE specialist (seed 1248); Square uses T2-C with fixed sparse-MoE candidates; Pinball uses T2-C with Deep-FNN-H3 candidates. The first archived valid window at the lowest held-out Reynolds number is shown at its endpoint: Circular, window 976, $t_0 = 1384.4960$, $\Delta T = 598.7009$, $K = 48$; Square H-P, window 506, $t_0 = 8$, $\Delta T = 96$, $K = 24$; Pinball H-P, window 2, $t_0 = 36.75$, $\Delta T = 14$, $K = 56$. Velocity panels show $\|\mathbf{u}\|_2$ and errors $\|\mathbf{u}_{pred} - \mathbf{u}_{ref}\|_2$; pressure panels show weighted-zero-mean kinematic pressure p° and $|p_{pred}^\circ - p_{ref}^\circ|$. Coordinates, time, and fields retain their native dataset scales; no additional cross-case nondimensionalization is imposed. In particular, the Circular source uses fixed $\nu = 0.001$. Each row uses a joint full-domain field range and a separate zero-based error range, without percentile clipping. The color scales are case-specific and support within-row comparison only. These snapshots are neither a matched architecture ablation nor evidence of one model’s zero-shot transfer between geometries.

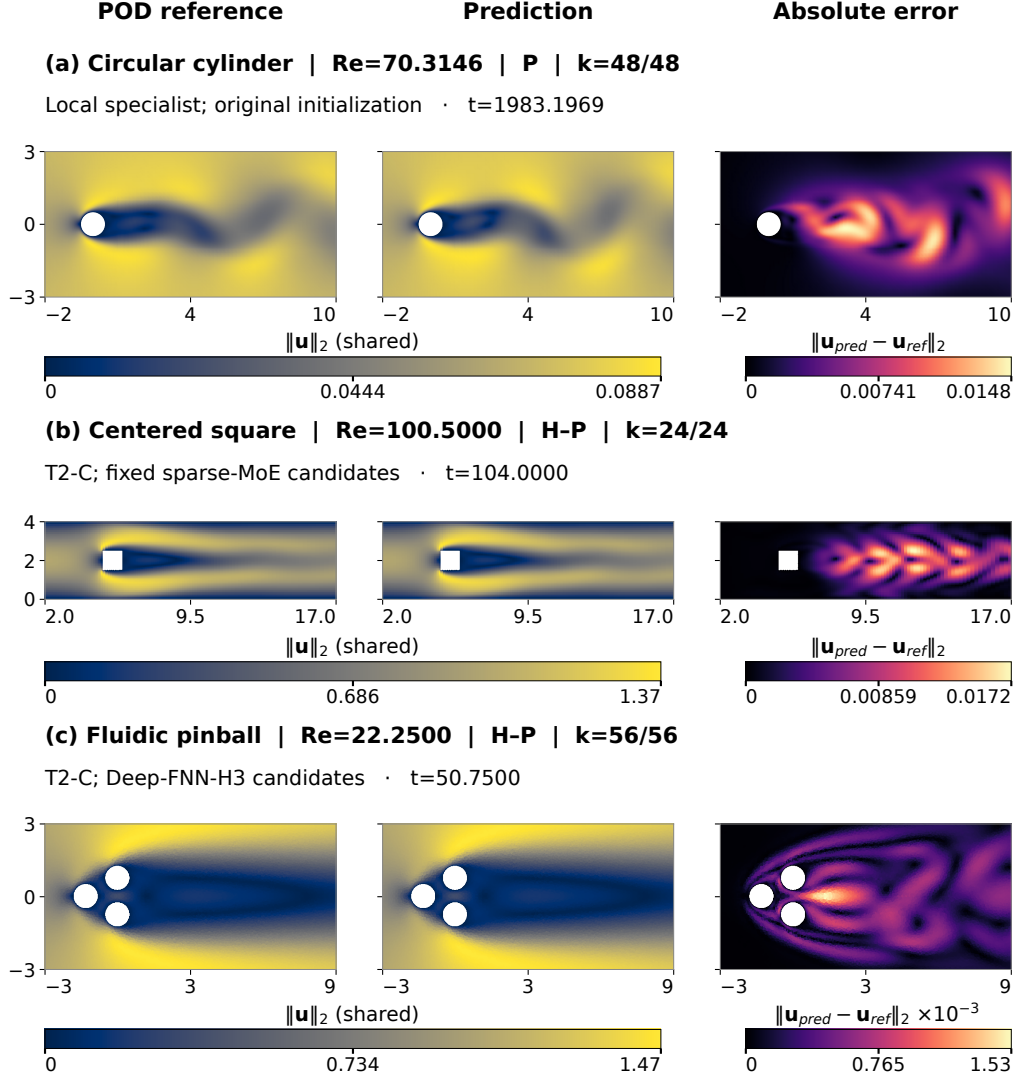


Figure 2: Representative velocity fields across three flow benchmarks. Columns show the POD-reconstructed reference, the case-specific prediction, and pointwise absolute error. Circular uses the original-initialization Periodic sparse-MoE specialist (seed 1248); Square uses T2-C with fixed sparse-MoE candidates; Pinball uses T2-C with Deep-FNN-H3 candidates. The first archived valid window at the lowest held-out Reynolds number is shown at its endpoint: Circular, window 976, $t_0 = 1384.4960$, $\Delta T = 598.7009$, $K = 48$; Square H-P, window 506, $t_0 = 8$, $\Delta T = 96$, $K = 24$; Pinball H-P, window 2, $t_0 = 36.75$, $\Delta T = 14$, $K = 56$. Velocity panels show $\|\mathbf{u}\|_2$ and errors $\|\mathbf{u}_{pred} - \mathbf{u}_{ref}\|_2$; pressure panels show weighted-zero-mean kinematic pressure p° and $|p^\circ_{pred} - p^\circ_{ref}|$. Coordinates, time, and fields retain their native dataset scales; no additional cross-case nondimensionalization is imposed. In particular, the Circular source uses fixed $\nu = 0.001$. Each row uses a joint full-domain field range and a separate zero-based error range, without percentile clipping. The color scales are case-specific and support within-row comparison only. These snapshots are neither a matched architecture ablation nor evidence of one model’s zero-shot transfer between geometries.

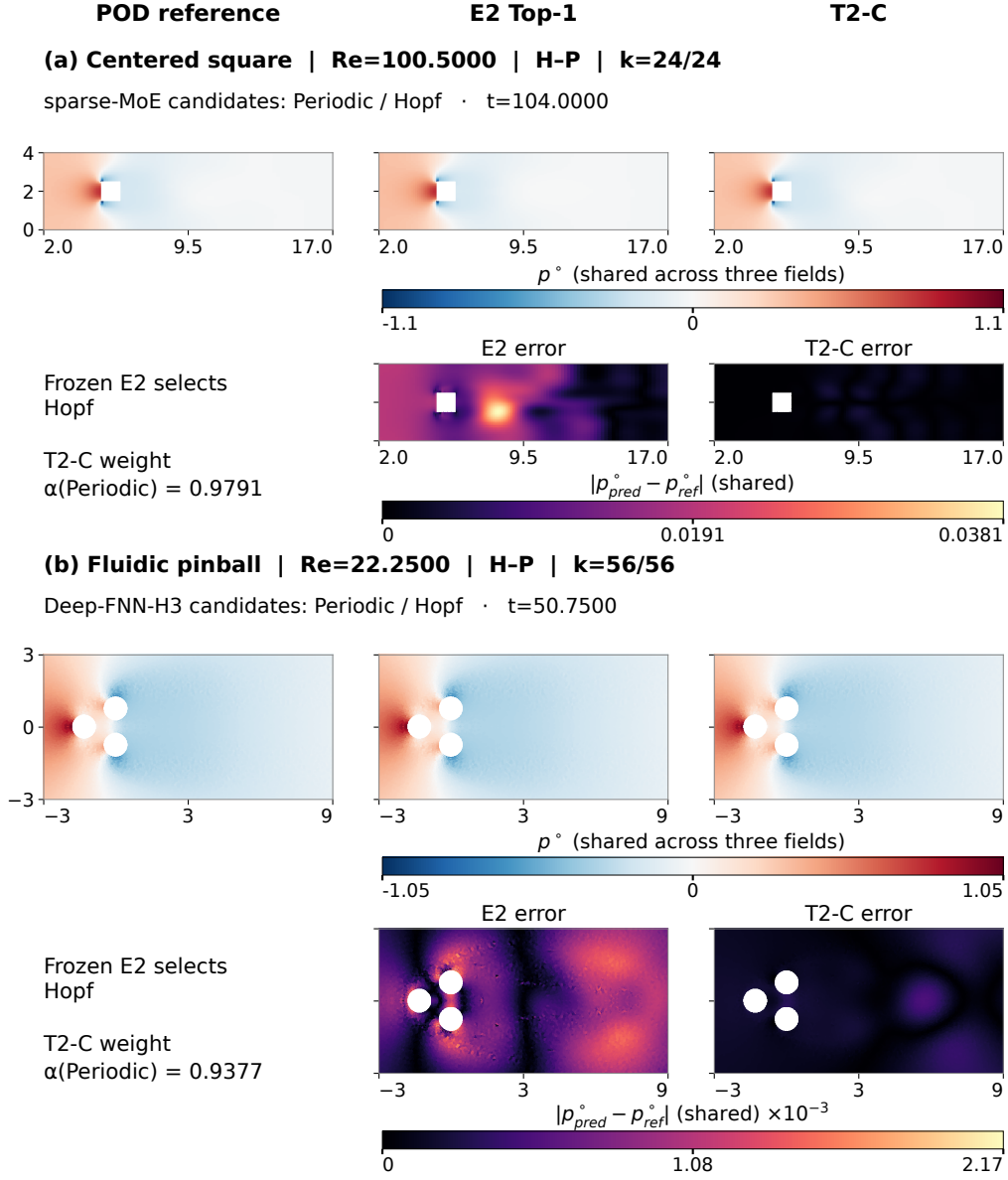


Figure 3: Frozen output-space fusion on held-out H-P windows, displaying pressure. Each case block shows the POD reference, the actual frozen E2 Top-1 prediction, and T2-C, followed by their two pointwise error fields. E2 is obtained by the global three-class router argmax; it selects a member of the prescribed pair in all shown windows, without using test errors. Square uses archived sparse-MoE candidates; Pinball uses Deep-FNN-H3 candidates. The displayed T2-C weight multiplies the first named candidate and is fixed from the initial history throughout the rollout; there is no fusion feedback. Velocity error is $\|\mathbf{u}_{pred} - \mathbf{u}_{ref}\|_2$; pressure error is $|p_{pred}^o - p_{ref}^o|$ after the same quadrature-weighted zero-mean gauge. All fields retain native dataset scales. Within each case, the three physical fields share one full-range scale and the two errors share one zero-based scale, with no clipping or smoothing. Case-specific scales are not comparable across rows. The first common valid window at the lowest held-out Re is shown at its endpoint, not selected by fusion benefit. Square: window 506, $t_0 = 8$, $\Delta t = 4$, $K = 24$, $t = 104$. Pinball: window 2, $t_0 = 36.75$, $\Delta t = 0.25$, $K = 56$, $t = 50.75$. Circular is not included in this H-P comparison because a matching verified H-P fusion field set was not established.

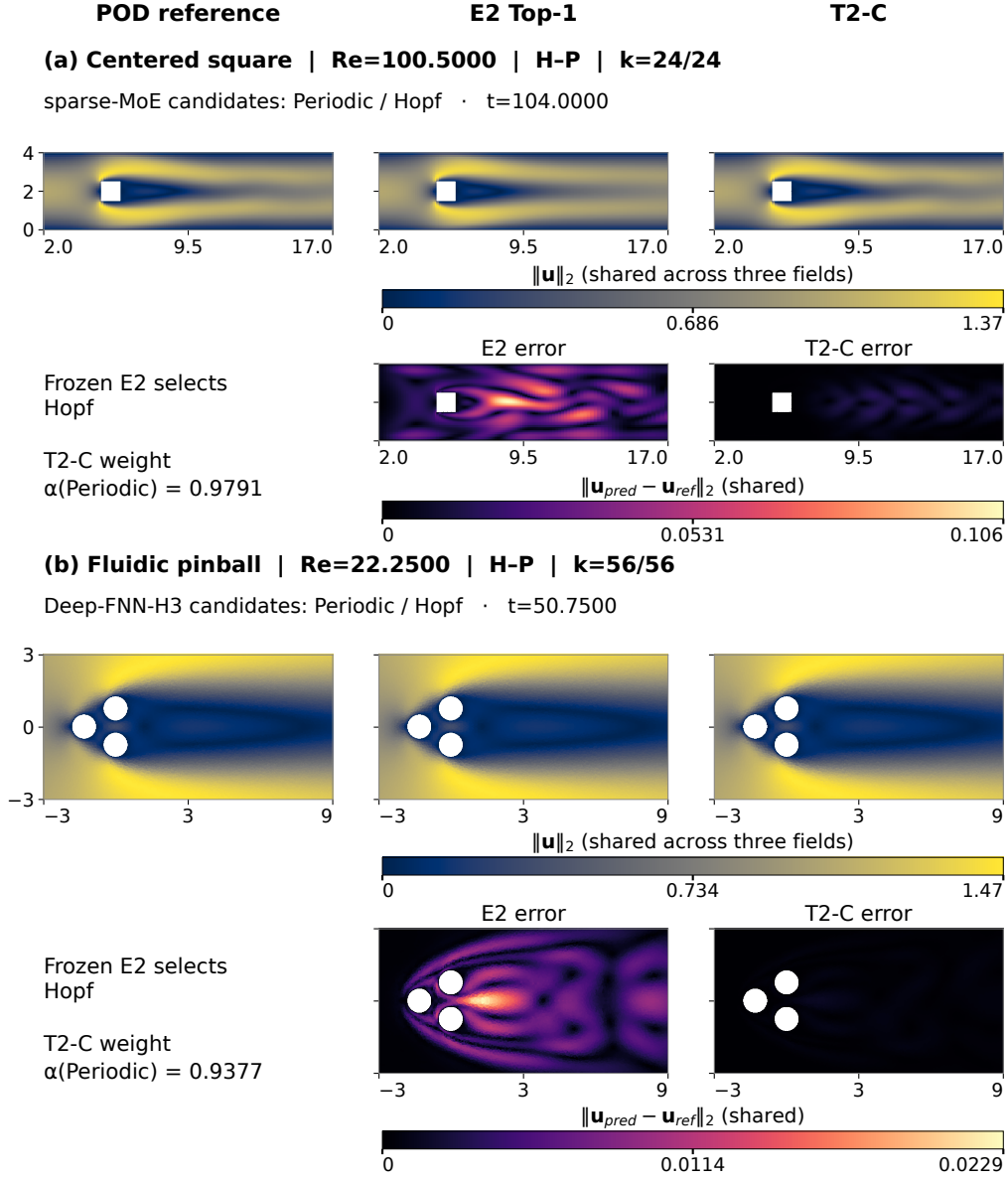


Figure 4: Frozen output-space fusion on held-out H-P windows, displaying velocity. Each case block shows the POD reference, the actual frozen E2 Top-1 prediction, and T2-C, followed by their two pointwise error fields. E2 is obtained by the global three-class router argmax; it selects a member of the prescribed pair in all shown windows, without using test errors. Square uses archived sparse-MoE candidates; Pinball uses Deep-FNN-H3 candidates. The displayed T2-C weight multiplies the first named candidate and is fixed from the initial history throughout the rollout; there is no fusion feedback. Velocity error is $\|\mathbf{u}_{pred} - \mathbf{u}_{ref}\|_2$; pressure error is $|p_{pred}^o - p_{ref}^o|$ after the same quadrature-weighted zero-mean gauge. All fields retain native dataset scales. Within each case, the three physical fields share one full-range scale and the two errors share one zero-based scale, with no clipping or smoothing. Case-specific scales are not comparable across rows. The first common valid window at the lowest held-out Re is shown at its endpoint, not selected by fusion benefit. Square: window 506, $t_0 = 8$, $\Delta t = 4$, $K = 24$, $t = 104$. Pinball: window 2, $t_0 = 36.75$, $\Delta t = 0.25$, $K = 56$, $t = 50.75$. Circular is not included in this H-P comparison because a matching verified H-P fusion field set was not established.

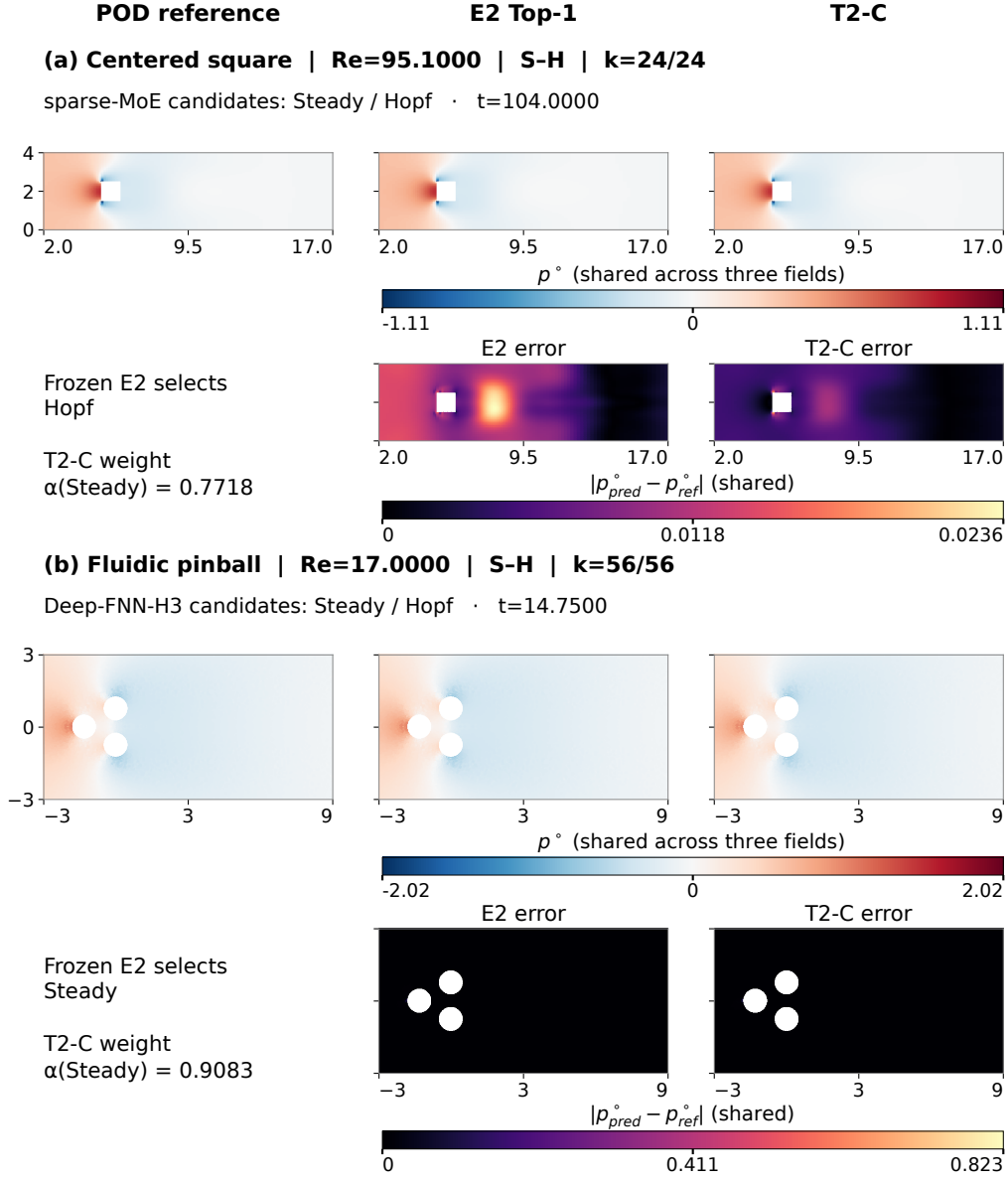


Figure 5: Frozen output-space fusion on held-out S–H windows, displaying pressure. Each case block shows the POD reference, the actual frozen E2 Top-1 prediction, and T2-C, followed by their two pointwise error fields. E2 is obtained by the global three-class router argmax; it selects a member of the prescribed pair in all shown windows, without using test errors. Square uses archived sparse-MoE candidates; Pinball uses Deep-FNN-H3 candidates. The displayed T2-C weight multiplies the first named candidate and is fixed from the initial history throughout the rollout; there is no fusion feedback. Velocity error is $\|\mathbf{u}_{pred} - \mathbf{u}_{ref}\|_2$; pressure error is $|p^o_{pred} - p^o_{ref}|$ after the same quadrature-weighted zero-mean gauge. All fields retain native dataset scales. Within each case, the three physical fields share one full-range scale and the two errors share one zero-based scale, with no clipping or smoothing. Case-specific scales are not comparable across rows. The first common valid window at the lowest held-out Re is shown at its endpoint, not selected by fusion benefit. Square: window 7814, $t_0 = 8$, $\Delta t = 4$, $K = 24$, $t = 104$. Pinball: window 514, $t_0 = 0.75$, $\Delta t = 0.25$, $K = 56$, $t = 14.75$. This supplement covers these two verified cases only; it is not a claim that Circular S–H fusion data do not exist.

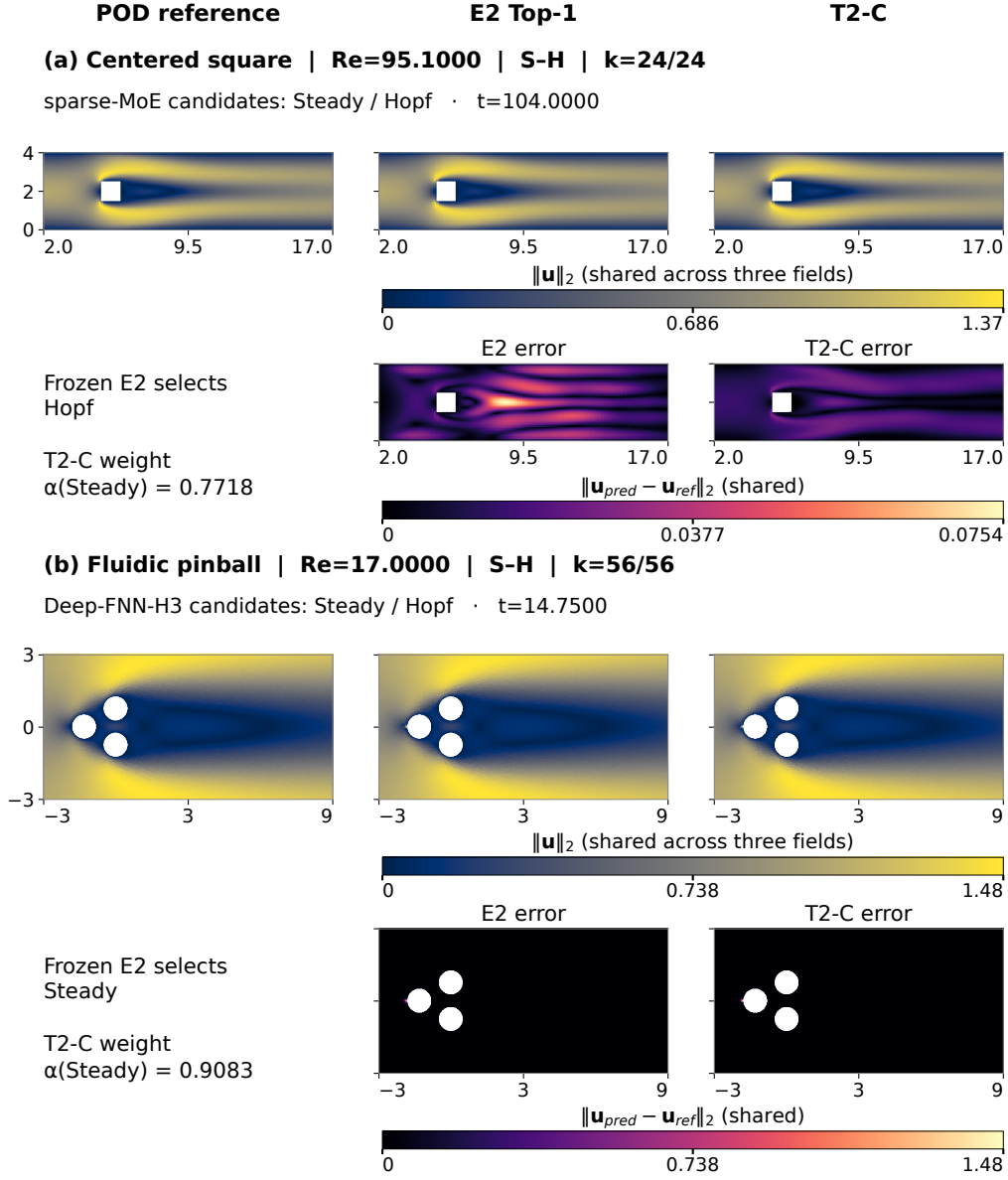


Figure 6: Frozen output-space fusion on held-out S–H windows, displaying velocity. Each case block shows the POD reference, the actual frozen E2 Top-1 prediction, and T2-C, followed by their two pointwise error fields. E2 is obtained by the global three-class router argmax ; it selects a member of the prescribed pair in all shown windows, without using test errors. Square uses archived sparse-MoE candidates; Pinball uses Deep-FNN-H3 candidates. The displayed T2-C weight multiplies the first named candidate and is fixed from the initial history throughout the rollout; there is no fusion feedback. Velocity error is $\|\mathbf{u}_{\text{pred}} - \mathbf{u}_{\text{ref}}\|_2$; pressure error is $|p_{\text{pred}}^{\circ} - p_{\text{ref}}^{\circ}|$ after the same quadrature-weighted zero-mean gauge. All fields retain native dataset scales. Within each case, the three physical fields share one full-range scale and the two errors share one zero-based scale, with no clipping or smoothing. Case-specific scales are not comparable across rows. The first common valid window at the lowest held-out Re is shown at its endpoint, not selected by fusion benefit. Square: window 7814, $t_0 = 8$, $\Delta t = 4$, $K = 24$, $t = 104$. Pinball: window 514, $t_0 = 0.75$, $\Delta t = 0.25$, $K = 56$, $t = 14.75$. This supplement covers these two verified cases only; it is not a claim that Circular S–H fusion data do not exist.